A Testing Schema for Methods and Systems of Methods

Supplement to *The Science of Methods - Register of the Elements of Method*

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Abstract

This paper presents a general, domain-independent testing schema for methods and systems of methods, derived from the Register. The schema integrates the algebraic substrate of Annex 0, the local–global organisation of Annex I, the operator-compatibility conditions of Annex II, the interface-alignment geometry of Annex III, and the evaluative continuation theory of Appendix B. A worked example in pure implicational natural deduction demonstrates the schema in a formal domain, and an addendum outlines an application to retrieval-augmented chat governance.

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1. Introduction

A method is a purposeful progression of steps in a domain, producing configurations that may be continued, compared, or evaluated. Systems of methods consist of arrangements in which multiple methods are combined, coordinated, or extended through one another.

The Register develops four formal layers for the representation and testing of methods and systems of methods: an algebraic layer (Annex 0), a sheaf-theoretic layer (Annex I), an operator-algebraic layer (Annex II), and a geometric layer (Annex III), together with evaluative continuation criteria (Appendix B).

This paper extracts those layers into a self-standing suite for testing methods and systems of methods. The schema is domain-independent in the sense of the Register: it applies in diverse domains to methods and systems whose behavior persists under variation of context and admits comparison under restriction and extension.

For such methods and systems, the schema identifies failure modes that cannot be detected by local behavior alone, including failures of continuation under domain expansion.

2. Primitive Objects (Annex 0)

Let D be a domain, an expanse in which elements may exist. A difference is any of the four kinds defined in Annex 0: between presence and absence of an element, between elements, between collections, or between an element and the rest of the domain. A relation is a difference.

A configuration is an arrangement of relations designated, ordered, or set aside for further study. A structure is a configuration designated for continuation.

A step is a deliberate act by which the practitioner alters a configuration of designated relations in response to one or more selected relations.

Let Σ be the set of available steps after the first retained difference. The free semigroup Σ^+ represents all finite step sequences.

A configuration map

$$E : \Sigma^+ \to \mathcal{R}$$

assigns to each step sequence its resulting configuration.

Two sequences are configuration-equivalent if

$$s \equiv s' \iff E(s) = E(s').$$

The quotient

$$Q = \Sigma^+ / \equiv$$

represents configurations observationally as equivalence classes of step sequences that produce the same configuration under E. This representation provides the object layer used in the tests that follow.

In the testing schema, configurations are compared through their classes in Q. References to configurations in the tests mean the corresponding equivalence classes unless stated otherwise.

3. Primary Operators

3.1 Update Operator (Continuation)

The update operator Φ is the partial map taking a configuration to the next configuration produced from it by continuation of the method.

3.2 Restriction Operator (Annex I)

For a subdomain $U \subseteq D$, let

$$R_U : Q \to Q_U$$

denote the induced restriction map. The value $R_U(q)$ is the configuration class obtained by restricting q to U. This restriction map removes all configuration content outside U.

4. Algebraic Compatibility (Annex II)

Compatibility of continuation with restriction concerns the interaction between the update operator Φ and the restriction operators.

Let $\rho_U : \mathcal{R} \to \mathcal{R}_U$ be the restriction of configurations to a subdomain U. The induced restriction on configuration classes is written

$$R_U([s]) := [s]_U,$$

where $[s]_U$ denotes the class in Q_U represented by

$$E_U(s) := \rho_U(E(s)).$$

Let Φ_U denote the update operator induced on the restricted domain U.

Compatibility of continuation with restriction is tested for configurations q in the relevant common domain by the operator condition:

$$R_U(\Phi(q)) = \Phi_U(R_U(q)).$$

Failure of this identity indicates incompatibility between continuation and restriction.

5. Local–Global Assembly Descent Test (Annex I)

The descent test proceeds by organizing the domain under study into subdomains, defining restriction maps among them, and identifying which families of subdomains count as covers. This establishes a site (Dom, J), where Dom consists of the subdomains together with the restriction maps among them, and J collects the families $\{U_i \to U\}$ taken as covers.

Let $\{U_i \to U\}$ be a covering family of the domain U. Local configurations $q_i \in Q_{U_i}$ agree on overlaps if

$$R_{U_i \cap U_j}(q_i) = R_{U_i \cap U_j}(q_j).$$

The local family $\{q_i\}$ satisfies descent if there exists a global configuration q such that

$$R_{U_i}(q) = q_i.$$

If no such q exists, the system exhibits a descent defect: local compatibility without global realizability.

6. Boundary Interfaces and Alignment (Annex III)

A method may consist of multiple configurations connected by interfaces. A boundary identification indicates, for a given interface, which parts of one configuration correspond to which parts of another where the configurations meet. A transition map specifies how the corresponding parts on one side of the interface are expressed in terms used on the other.

Let $\mathcal{I}$ denote the set of boundary interfaces between configurations.

For each interface $e \in \mathcal{I}$, the practitioner assigns an alignment map whose values represent the practitioner's assessment of the compatibility of configurations at that interface.

The geometric alignment functional aggregates the local interface values produced by the Annex III alignment maps into a single global score. For a finite interface set $\mathcal{I}$, the schema records the unweighted mean alignment of interfaces as the finite special case of the Annex III global alignment functional:

$$A_{\mathrm{geom}}(M) = \frac{1}{|\mathcal{I}|} \sum_{e \in \mathcal{I}} a_e(M),$$

where $a_e(M)$ denotes the alignment value assigned by the practitioner at interface e for the method application being evaluated.

Instability appears when this global alignment decreases under continuation.

7. Evaluative Interfaces (Appendix B)

7.1 Evaluative Regime and Interfaces

The testing schema distinguishes formal characteristics inherent in a method itself from evaluative quantities assigned by the practitioner within a testing regime. These assignments fix the reference outputs, admissibility criteria, and tolerance levels under which method outputs are tested within a particular domain. By making those assignments explicit, the schema renders the criteria of assessment inspectable and comparable in different applications.

In the broader vocabulary of the Register, plausibility governs the proposal of a further step where veracity is not yet settled, while coherence is the practitioner's determination that a configuration is sufficient for continuation. Following Appendix B, that determination is operationalised through the explicit evaluative quantities retained below.

In Appendix B, evaluative interfaces are the elements of an evaluation set at which method outputs are tested. This use differs from the boundary-interface terminology of Annex III, where an interface denotes a relation between connected configurations.

In the test schema, evaluative interfaces are indexed by configuration classes $e \in Q$, and the method output assessed at an evaluative interface e is written $y_e(M)$.

7.2 Local Evaluative Quantities

Each evaluative interface is associated with a reference output y_e^{ref} designated by the practitioner within the testing regime for the domain under study. This reference output determines the reference condition used in the veracity test.

For each evaluative interface e, define the local quantities:

- **Alignment** $a_e(M)$, assigned through an alignment map comparing the method output $y_e(M)$ with the reference output y_e^{ref};
- **Veracity** $v_e(M)$, indicating whether, in the judgment of the practitioner, the method output satisfies the reference condition at e;
- **Uncertainty** $u_e(M)$, representing the variability of the method output under repeated evaluation at the same interface.

These quantities correspond to the alignment, veracity, and uncertainty maps defined in Appendix B.

Appendix B presupposes sufficient intensity of application for these quantities to be defined at all. Where intensity falls below the minimal detection threshold, the descriptive quantities are undefined. Intensity is therefore treated as a background condition of evaluability rather than as a separate explicit variable in the local test.

The schema combines these quantities into a local acceptance test at each evaluative interface. The local test operator records the practitioner's determination that the configuration at the interface is coherent enough for continuation under the admissibility conditions of the testing regime.

A local test operator is

$$T_{\text{loc}}(M, e) = 1 \quad \text{iff} \quad a_e(M) \geq a_{\min},\ v_e(M) \geq v_{\min},\ u_e(M) \leq u_{\max}.$$

$T_{\text{loc}}(M, e) = 1$ signifies that the method passes the local test at interface e, while $T_{\text{loc}}(M, e) = 0$ indicates failure of the local test.

The practitioner assigns thresholds $a_{\min}, v_{\min}, u_{\max}$ representing the minimum acceptable alignment and veracity and the maximum tolerated uncertainty at an evaluative interface. Their values express the tolerance levels adopted in the testing regime for the domain under study.

8. Composite Tests (Appendix B)

Appendix B includes composite tests for methods and systems of methods whose outputs must remain jointly compatible over families of evaluative interfaces. Composite testing records whether outputs that pass local testing also remain compatible when taken together.

Let $\mathcal{F}$ denote a family of finite subsets of evaluative interfaces selected by the practitioner for composite assessment. For each $F \in \mathcal{F}$, the practitioner assigns a composite compatibility value $c_F(M)$ representing the compatibility of the outputs produced at the interfaces in F.

Fix a composite threshold $c_{\min}$. Define the composite test operator

$$T_{\text{comp}}(M, F) = 1 \quad \text{iff} \quad c_F(M) \geq c_{\min} \quad \text{and} \quad T_{\text{loc}}(M, e) = 1 \text{ for all } e \in F.$$

Composite failure records that outputs acceptable at each interface separately do not remain acceptable when taken together. Composite testing is required whenever continuation depends on the joint compatibility of multiple evaluated outputs.

9. Global Alignment Functional (Appendix B)

The geometric alignment functional $A_{\text{geom}}(M)$ of Section 6 records compatibility at boundary interfaces, while the global alignment functional $\mathcal{C}$ defined is the Appendix-B evaluative aggregate over alignment, veracity, and uncertainty on the selected evaluation set.

Let $\mathcal{E}$ be an evaluation set consisting of evaluative interfaces selected by the practitioner for evaluation of the method or system under study.

The practitioner selects coefficients λ_A, λ_V, and λ_U in the formula below to set the relative weight assigned to alignment, veracity, and uncertainty within the testing regime. In different domains these coefficients may be weighted differently depending on the role each dimension plays in assessing continuation.

Define

$$\mathcal{C}[M \mid \mathcal{E}] = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} (\lambda_A a_e(M) + \lambda_V v_e(M) - \lambda_U u_e(M)).$$

10. Restricted and Broadened Evaluation Sets

Let $\tau_{\text{low}} < \tau_{\text{high}}$ be continuation evaluation thresholds chosen by the practitioner to separate stable, unstable, and failure regimes of the method. Values of the global alignment functional $\mathcal{C}[M \mid \mathcal{E}]$ above τ_{high} indicate stable continuation, while values below τ_{low} indicate failure of continuation.

Let

$$\mathcal{E}_1 \subset \mathcal{E}_2$$

denote evaluation sets selected by the practitioner under restricted and broadened evaluation regimes, respectively.

A method exhibits overfit under broadening if

$$\mathcal{C}[M \mid \mathcal{E}_1] \geq \tau_{\text{high}} \quad \text{and} \quad \mathcal{C}[M \mid \mathcal{E}_2] < \tau_{\text{low}}.$$

This is the signature of continuation collapse.

11. Corrective Measures and Effective Correction (Appendix B)

Appendix B treats evaluative outcomes not only as classifications of method performance but also as grounds for method revision. Let $\mathcal{R}_t$ denote the accumulated history of local, composite, and global test outcomes up to stage t.

In Section 3, Φ denotes the update operator on configurations. In this section, following Appendix B, Φ denotes the correction operator acting on methods under accumulated test history.

The corrective step is written

$$M_{t+1} = \Phi(M_t, \mathcal{R}_t).$$

A correction cycle is effective if

$$\mathcal{C}[M_{t+1} \mid \mathcal{E}_2] \geq \mathcal{C}[M_t \mid \mathcal{E}_2]$$

and the local and composite test conditions remain satisfied.

The preceding conditions govern a single correction step. Iterative correction concerns the behavior of a history-dependent sequence of such steps.

Iterative correction is effective as a history-dependent process when the correction operator

$$\Phi(M_t, \mathcal{R}_t)$$

depends on the accumulated test history $\mathcal{R}_t$, and when the resulting sequence

$$M_0, M_1, M_2, \ldots$$

contains some stage t such that

$$\mathcal{C}[M_{t+1} \mid \mathcal{E}_2] > \mathcal{C}[M_t \mid \mathcal{E}_2],$$

while the local and composite test conditions remain satisfied.

Iterative correction operates by design in adaptive methods, such as feedback or PID controllers, but can also be used more generally where outcomes of application are retained and used to revise the governing rule or procedure of later applications of a method.

12. Generative Continuation under Correction

Let M_t be the method after t correction cycles.

Successive correction cycles generate the sequence

$$M_0, M_1, M_2, \ldots$$

Section 11 establishes when a single correction cycle is effective and when iterative correction is useful as a history-dependent process. Generative continuation under correction adds recovery and maintenance of stable continuation under broadened evaluation.

A method exhibits generative continuation if

$$\exists T : \ \forall t \geq T, \ \mathcal{C}[M_t \mid \mathcal{E}_2] \geq \tau_{\text{high}}.$$

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13. Worked Example: $\to$-Only Natural Deduction

This example highlights the evaluative continuation apparatus of Appendix B in a purely formal setting. The method performs correctly on restricted proof tasks but fails under broadened evaluation when nested discharges are introduced. The example demonstrates instability under evaluation broadening and the role of corrective iteration in a formal domain.

In natural deduction, a proof is built step by step from assumptions by applying rules of inference. The implication-only fragment uses just one connective, $\to$, so every proof task asks whether a formula involving implication can be derived from a given set of assumptions. Two rules are central. Implication elimination allows one to infer ψ from φ and $\varphi \to \psi$. Implication introduction allows one to prove $\varphi \to \psi$ by temporarily assuming φ, deriving ψ under that temporary assumption, and then discharging the assumption. A discharge is the closure of a temporary assumption once the corresponding implication has been established. The example below concerns proof tasks in which such discharges are nested inside one another. The local rule applications may remain correct at each stage, while broadened evaluation reveals instability in the handling of those nested dependencies.

Purpose

A testing specification for a generative method M operating on a purely formal domain D. The specification implements Appendix B's local tests, composite tests, global alignment functional $\mathcal{C}$, and the overfit diagnostic pattern under restricted and broadened evaluation sets.

13.1 Formal Domain D

The domain D consists of proof tasks and the configurations arising in their development within the implication-only fragment of propositional natural deduction.

Language. Let $\mathsf{Var} = \{p, q, r, \dots\}$ be propositional atoms. Formulas are generated by:

$$\varphi ::= x \mid (\varphi \to \psi) \qquad (x \in \mathsf{Var}).$$

Define implication depth depth$(\cdot)$ by:

$$\mathrm{depth}(x) = 0, \qquad \mathrm{depth}(\varphi \to \psi) = 1 + \max(\mathrm{depth}(\varphi), \mathrm{depth}(\psi)).$$

Judgments (Goals). A proof task is a sequent: $\Gamma \vdash \varphi$. This means that the formula φ is to be derived from the assumptions collected in Γ.

Calculus. The implication-only natural-deduction calculus uses three forms of step: assumption, which opens a temporary hypothesis; implication introduction, which discharges such a hypothesis in order to establish an implication; and implication elimination, which derives the consequent of an implication from the implication together with its antecedent.

13.2 Configurations and Evaluative Interfaces

Configuration. A configuration is represented by the tuple:

$$\sigma := (\Gamma,\ g,\ \Delta,\ \Pi),$$

where Γ is the current context of available assumptions, g is the current goal formula, Δ is the set of judgments already derived, and Π is the stack of open temporary assumptions that have not yet been discharged.

Evaluative Interfaces. An evaluative interface e is a configuration selected by the practitioner for assessment.

Let $\mathcal{E}$ denote the set of evaluative interfaces selected by the practitioner for testing.

Output. At each interface e, the method produces an output $y_e(M)$ consisting of a next proof step, such as opening an assumption, applying implication elimination, or discharging an assumption by implication introduction.

13.3 Reference Output and Local Quantities

Reference Output. The reference output at e is a selected rule-valid application of the $\to$-natural-deduction calculus. Rule-valid means that the step is licensed by one of the three rules of the calculus and is correctly applicable to the configuration at hand. In this example, the reference condition is satisfaction of the rule exhibited by that reference output.

$$a_e(M) = \begin{cases} 1 & \text{if the step is rule-valid,} \\ 0 & \text{otherwise,} \end{cases}$$

$$v_e(M) = \begin{cases} 1 & \text{if the step satisfies the reference rule,} \\ 0 & \text{otherwise.} \end{cases}$$

Uncertainty Map (Repeatability). Fix $N \geq 2$. Re-run M at evaluative interface e exactly N times. Let $q_e(M)$ be the proportion of valid trials.

$$u_e(M) := 1 - q_e(M).$$

This defines uncertainty as instability of continuation.

13.4 Local Test

Fix strict thresholds

$$a_{\min} = 1, \qquad v_{\min} = 1, \qquad u_{\max} = 0.$$

Define the local test operator

$$T_{\mathrm{loc}}(M,e) = \begin{cases} 1 & \text{if } a_e(M) \geq a_{\min},\ v_e(M) \geq v_{\min},\ u_e(M) \leq u_{\max} \\ 0 & \text{otherwise.} \end{cases}$$

In the restricted tasks considered, the output at each evaluative interface is rule-valid, satisfies the reference rule, and is repeatable under re-evaluation. At each such interface,

$$a_e(M) = 1, \qquad v_e(M) = 1, \qquad u_e(M) = 0,$$

and the local test takes the value

$$T_{\mathrm{loc}}(M,e) = 1.$$

13.5 Composite Test

The composite test asks whether outputs produced at a finite family of evaluative interfaces remain jointly compatible when taken together within a proof development.

For a finite family F of evaluative interfaces drawn from a proof task, define

$$T_{\mathrm{comp}}(M,F) = 1$$

when the outputs at the interfaces in F are jointly compatible and the corresponding local tests are all satisfied.

The rule-valid outputs remain jointly compatible when taken together in the proof development, meaning that they can occur together in a single derivation without conflict among assumptions, goals, or discharges, and the corresponding local tests are satisfied throughout. The composite test takes the value

$$T_{\mathrm{comp}}(M,F) = 1.$$

13.6 Global Alignment Functional

Weights: $\lambda_A = \lambda_V = \lambda_U = 1$.

$$\mathcal{C}[M \mid \mathcal{E}] = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} (a_e(M) + v_e(M) - u_e(M)).$$

13.7 Evaluation Sets

The practitioner forms evaluation sets by selecting evaluative interfaces arising in the proof tasks under consideration.

A discharge is the closure of an open assumption by an application of the $\to$-Introduction rule. A discharge is nested when such a closure occurs within a proof task that still depends on other open assumptions.

Two evaluation regimes are considered. The restricted regime uses

$$\mathcal{E}_1,$$

consisting of evaluative interfaces selected from proof tasks of depth at most 2 with at most one open assumption. The broadened regime uses

$$\mathcal{E}_2,$$

consisting of evaluative interfaces selected from proof tasks that exceed those restrictions, including proof tasks involving nested discharges.

13.8 Restricted vs Broadened Sets

Task Suites.

- S_1 (Restricted): Goals with depth ≤ 2, with at most one open assumption (e.g., $\vdash p \to p$).
- S_2 (Broadened): Goals of greater depth, including proof tasks with multiple nested discharges.

13.9 The Failure Pattern (Discharge-Instability)

Fix continuation-evaluation thresholds

$$\tau_{\text{low}} = 1.5, \qquad \tau_{\text{high}} = 1.9.$$

Values of the global alignment functional above τ_{high} indicate stable continuation, while values below τ_{low} indicate continuation failure under broadened evaluation.

Assume method M works perfectly on simple tasks (S_1) but exhibits instability in proof tasks with nested discharges (S_2).

- On $\mathcal{E}_1$ (selected by the practitioner from configurations arising in S_1): $a_e(M) = 1, v_e(M) = 1, u_e(M) = 0$. Thus $\mathcal{C}[M \mid \mathcal{E}_1] = 2 \geq 1.9$.
- On $\mathcal{E}_2$ (selected by the practitioner from configurations arising in S_2): A fraction α of evaluative interfaces are unstable ($q_e = 0.6 \implies u_e(M) = 0.4,\ a_e(M) = 0,\ v_e(M) = 0$).

Numerical Representation:

$$\mathcal{C}[M \mid \mathcal{E}_2] = 2(1-\alpha) - 0.4\alpha = 2 - 2.4\alpha.$$

Failure below $\tau_{\text{low}} = 1.5$ occurs when $2 - 2.4\alpha < 1.5$, which implies $\alpha > 20.8\%$.

13.10 Summary of Diagnostic Outcome

Overfit is witnessed when

$$\mathcal{C}[M \mid \mathcal{E}_1] \geq \tau_{\text{high}}$$

while

$$\mathcal{C}[M \mid \mathcal{E}_2] < \tau_{\text{low}}.$$

The instability appears at evaluative interfaces arising in proof tasks with nested discharges, where repeatability fails. Local rule validity is insufficient to guarantee continuation under broadened evaluation.

14. Test Suite Summary

The testing schema evaluates methods and systems of methods in six distinct dimensions:

1. Configuration equivalence (Annex 0)

2. Algebraic compatibility (Annex II)
3. Local–global assembly (Annex I)
4. Composite testing (Appendix B)
5. Boundary-interface alignment (Annex III)
6. Evaluative continuation (Appendix B)

Depending on the method or system under study, some dimensions may not apply, while others diagnose failure of continuation, operator incompatibility, or instability under broadened evaluation.

15. Conclusion

The test schema is portable and capable of detecting failure modes that elude purely local analysis. The worked example and the addendum demonstrate that continuation demands more than local compatibility, and that generative instability may remain undetected without the evaluative continuation criteria of Appendix B.

Addendum: Prospective Application to Retrieval-Augmented Chat Governance

This addendum outlines a prospective application of the schema to a chat-type language model that answers user queries by combining the current conversation state with retrieved text passages. It describes a governance model for evaluation and corrective revision of a chat process.

Purpose

A prospective governance specification for a generative conversational method M operating on a dialogue-and-retrieval domain D. The specification implements Annex II compatibility, Annex I local–global assembly, Annex III interface alignment, Appendix B local and composite tests, the global alignment functional $\mathcal{C}$, the overfit diagnostic pattern between restricted and broadened evaluation, and a model of iterative correction.

A.1 Formal Domain D

The domain D consists of multi-turn user queries together with retrieved text passages selected for response construction.

A conversational instance consists of a current user query, a prior turn history, a finite retrieved evidence set, and a candidate response.

A conversational instance is solvable if it admits a response that is relevant to the query and faithful to the retrieved evidence.

A.2 Configurations, Update Operator, and Evaluative Interfaces

Configuration. A configuration is a conversation state together with its active evidence set, the current candidate claim proposed by the language model generator, and the admission state under which that candidate is either accepted or rejected.

Update Operator. The update operator Φ is the partial map taking a configuration to the next configuration produced by one admissible response-construction step. Each step proceeds through a determination loop: the generator proposes a candidate claim or response step, the admission procedure tests that candidate for admissibility and compatibility, and the resulting decision is either accepted into the response state or rejected. The update consists of one such accepted step together with any associated revision of the response draft.

Restriction. For a subdomain $U \subseteq D$, restriction sends a conversational configuration to the corresponding reduced state on U, for example by limiting the turn history or narrowing the evidence set.

Evaluative Interfaces. An evaluative interface e is a response point at which the method must decide whether to include, exclude, or revise a claim in light of the evidence and the current conversational state.

Let $\mathcal{E}$ denote the evaluative interfaces designated by the governance protocol for testing.

Output. At each interface e, the method produces an output $y_e(M)$ consisting of a proposed claim or response step.

A.3 Reference Output and Local Quantities

Reference Output. The reference output at an evaluative interface is the claim or response step that preserves orientation toward the conversational purpose. In this example, the reference condition is satisfaction at the interface of relevance to the query and fidelity to the retrieved evidence.

Plausibility characterizes the generator's proposal of a candidate claim before full assessment is settled. Alignment, veracity, and uncertainty enter into the determination of whether the resulting conversational configuration is sufficient for continuation.

Define the local quantities

$$a_e(M) = \begin{cases} 1 & \text{if the output is aligned with the retrieved evidence,} \\ & \text{prior turns, and retained response state at } e, \\ 0 & \text{otherwise,} \end{cases}$$

$$v_e(M) = \begin{cases} 1 & \text{if the output preserves orientation toward the} \\ & \text{conversational purpose at } e, \\ 0 & \text{otherwise.} \end{cases}$$

Uncertainty Map (Repeatability). Fix $N \geq 2$. Re-run M at evaluative interface e exactly N times under the same conversational and evidence conditions. Let $q_e(M)$ be the proportion of trials yielding the same output and the same local result. Define

$$u_e(M) := 1 - q_e(M).$$

This defines uncertainty as instability of continuation at the interface.

At an evaluative interface, the schema records whether the current conversational configuration is sufficient for continuation under the stated local conditions. This sufficiency is determined from veracity, alignment, and uncertainty at the interface, relative to the retained conversational state and the conversational purpose.

A.4 Local Test

Fix strict local thresholds

$$a_{\min} = 1, \qquad v_{\min} = 1, \qquad u_{\max} = 0.$$

Define the local test operator

$$T_{\mathrm{loc}}(M, e) = \begin{cases} 1 & \text{if } a_e(M) \geq a_{\min},\ v_e(M) \geq v_{\min},\ u_e(M) \leq u_{\max}, \\ 0 & \text{otherwise.} \end{cases}$$

In the restricted conversational tasks considered, the relevant evidence is short and directly matched to the user query. At each such interface, the generator proposes a plausible claim, and the admission procedure yields a configuration satisfying the stated local conditions for continuation. The claim remains aligned with the evidence and oriented toward the conversational purpose, and the same result recurs under repeated evaluation. At each such interface,

$$a_e(M) = 1, \qquad v_e(M) = 1, \qquad u_e(M) = 0,$$

and the local test takes the value

$$T_{\mathrm{loc}}(M, e) = 1.$$

A.5 Composite Test

The composite test asks whether outputs produced at a finite family of evaluative interfaces remain jointly compatible when taken together within the same response.

For a finite family F of evaluative interfaces designated by the governance protocol, define

$$T_{\mathrm{comp}}(M, F) = 1$$

when the outputs at the interfaces in F remain jointly compatible and the corresponding local tests are all satisfied.

Joint compatibility means that the admitted claims remain mutually compatible when taken together as one response, and that the resulting response remains aligned with the evidence set as a whole and oriented toward the conversational purpose.

In the restricted conversational tasks, locally acceptable claims remain jointly compatible when taken together, and the corresponding local tests are satisfied throughout. The composite test takes the value

$$T_{\text{comp}}(M, F) = 1.$$

A.6 Algebraic Compatibility (Annex II)

Let U be a restricted conversational subdomain. Compatibility of continuation with restriction is tested by

$$R_U(\Phi(q)) = \Phi_U(R_U(q)).$$

In this example, the identity may fail. A response-construction step taken in the full conversational state may depend on evidence passages or prior-turn constraints that are removed by restriction. Restricting first can alter the next admissible response step. The method may exhibit an Annex II defect whenever continuation depends on information lost under restriction.

A.7 Local–Global Assembly (Annex I)

Local response fragments may agree on overlaps while still failing to assemble into a globally sufficient answer.

A response state is globally sufficient if its admitted claims remain aligned with the relevant evidence, preserve orientation toward the conversational purpose, and satisfy the conditions for continuation under broadened evaluation.

A.8 Boundary Interfaces and Alignment (Annex III)

Boundary interfaces occur where user query, retrieved passages, prior turns, and candidate response segments meet, and where generator output meets the admission procedure within the determination loop. The governance protocol assigns alignment values to these interfaces according to whether the adjacent components remain mutually compatible.

For a finite interface set $\mathcal{I}$,

$$A_{\text{geom}}(M) = \frac{1}{|\mathcal{I}|} \sum_{e \in \mathcal{I}} a_e^{\text{geom}}(M).$$

In the restricted conversational tasks, alignment remains high because the query, evidence, and admitted claims fit together. Under broadened evaluation, alignment may decrease when retrieved passages are partial or differently scoped.

A.9 Global Alignment Functional

Weights: $\lambda_A = \lambda_V = \lambda_U = 1$.

$$\mathcal{C}[M \mid \mathcal{E}] = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} (a_e(M) + v_e(M) - u_e(M)).$$

On the restricted set $\mathcal{E}_1$, each selected interface lies in a conversational instance where plausible candidate claims remain aligned with the evidence, veracity remains oriented toward the conversational purpose, and the admitted response state satisfies the stated local conditions for continuation, so

$$\mathcal{C}[M \mid \mathcal{E}_1] = 2.$$

On the broadened set $\mathcal{E}_2$, let α be the instability density, that is, the fraction of evaluative interfaces at which the generator produces a plausible claim that fails the local conditions of the full conversational state. At those interfaces, the output remains aligned with selected evidence and so $a_e(M) = 1$, but orientation toward the full conversational purpose fails and so $v_e(M) = 0$, while repeatability remains stable and so $u_e(M) = 0$. On the remaining interfaces,

$$a_e(M) = 1, \qquad v_e(M) = 1, \qquad u_e(M) = 0.$$

The global alignment functional on the broadened set is

$$\mathcal{C}[M \mid \mathcal{E}_2] = 2 - \alpha.$$

A.10 Evaluation Sets

The governance protocol fixes evaluation sets by designating evaluative interfaces from conversational tasks under consideration.

Two evaluation regimes are considered. The restricted regime uses

$$\mathcal{E}_1,$$

consisting of evaluative interfaces drawn from short factual queries with tightly matched retrieved evidence and minimal conversational ambiguity.

The broadened regime uses

$$\mathcal{E}_2,$$

consisting of evaluative interfaces drawn from longer, or cross-turn conversational tasks in which retrieved evidence may be partial or differently scoped.

A.11 Restricted vs Broadened Sets

Task Classes.

- S_1 (Restricted): short factual or explanatory tasks in which retrieved evidence is consistent and sufficient.
- S_2 (Broadened): longer or more ambiguous tasks in which the evidence set is incomplete, internally divergent, or distributed over multiple turns and passages.

A.12 The Failure Pattern (Context-Integration Overfit)

Fix continuation-evaluation thresholds

$$\tau_{\text{low}} = 1.5, \qquad \tau_{\text{high}} = 1.9.$$

Values of the global alignment functional above τ_{high} indicate stable continuation, while values below τ_{low} indicate continuation failure under broadened evaluation.

On $\mathcal{E}_1$,

$$\mathcal{C}[M \mid \mathcal{E}_1] = 2 \geq \tau_{\text{high}}.$$

On $\mathcal{E}_2$,

$$\mathcal{C}[M \mid \mathcal{E}_2] = 2 - \alpha.$$

Failure below $\tau_{\text{low}} = 1.5$ occurs when

$$2 - \alpha < 1.5,$$

which implies

$$\alpha > 0.5.$$

The method exhibits overfit under broadening when the instability density exceeds one half: plausible claims continue to be generated and accepted, while the resulting conversational configurations no longer satisfy the stated local conditions under the broadened regime.

A.13 Corrective Measures and Generative Continuation

The source of failure lies in the rule governing evidence integration and claim admission: local plausibility is treated as sufficient ground for inclusion in the final response without a sufficiently strong admission test under the stated local conditions.

A corrective measure revises the evidence-integration and admission rules so that plausible local fit with one passage is no longer by itself sufficient for claim admission.

Let M_t denote the conversational method after t correction cycles, and let α_t denote the fraction of broadened evaluative interfaces at stage t at which the current integration rule still admits a locally plausible but globally inadequate claim.

Then

$$\mathcal{C}[M_t \mid \mathcal{E}_2] = 2 - \alpha_t.$$

A correction cycle is effective if

$$\alpha_{t+1} < \alpha_t,$$

so that

$$\mathcal{C}[M_{t+1} \mid \mathcal{E}_2] > \mathcal{C}[M_t \mid \mathcal{E}_2],$$

while the local and composite test conditions remain satisfied.

Generative continuation under correction is recovered once

$$2 - \alpha_t \geq \tau_{\text{high}},$$

that is,

$$\alpha_t \leq 0.1.$$

In this prospective application, outcomes of broadened conversational failure are retained, and the evidence-integration and admission rules are revised in light of that history. Successive revisions applying the corrective measures of the schema can restore continuation.